Gender Hiring Quotas

Valentina D'Anna* Eliana Sena†

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Abstract

Gender quotas have mostly targeted corporate boards and done little for the gender employment gap. Requiring a minimum share of women in a firm's entire workforce would force firms to lay off workers or hire beyond their needs. A quota on new hires avoids these costs, but firms can comply and then undo its effect through separations. We study a novel type of quota: Italian firms that win public contracts must reserve for women at least 30 percent of the new hires needed for the project. Coverage depends on how contracts are funded, so similar contracts are awarded with and without the quota. Linking procurement records to employer-employee data and balance sheets, we compare winners of contracts with and without the quota in a matched difference-in-differences design. Three years after the award, female employment in firms subject to the quota is 11 percent higher than in comparable winners without it, with no difference for men, although some firms avoid the requirement by hiring less. Female separations rise two years after the award, but female employment remains above its pre-award level. The new jobs are mostly white-collar but temporary and part-time, with no effect on wages. We find no evidence of efficiency losses. Hiring quotas can bring women into male-dominated firms, but their effects partly fade once the obligation expires.

Keywords: Gender hiring quotas; Public procurement; Affirmative action; Labor demand; Female employment; Matched employer-employee data.

JEL Codes: H57, J16, J23, J78.

*Rome Economics Doctorate (RED): EIEF, LUISS, Tor Vergata, and Sapienza.

†University of Pennsylvania.

1 Introduction

Despite decades of progress, gender gaps persist both in employment and in top positions (Olivetti and Petrongolo, 2016; Blau and Kahn, 2017). In OECD countries, 72.5 percent of women are employed, against 87.7 percent of men.[1] To close the gap at the top, many countries have introduced gender quotas for corporate boards.[2] These quotas have raised women's presence on boards, but they reach a few, highly selected women. They can narrow the employment gap only indirectly, through spillovers to women below the top, and studies find little such effect (Bertrand et al., 2019; Maida and Weber, 2022). Acting on the employment gap directly through a quota on the stock of employment would force firms either to break existing matches or to hire beyond their labor demand. A quota on new hires avoids both problems: it can nudge firms toward a more gender-balanced workforce as they replace and expand their staff. Whether it works depends on three questions. First, is the effect long lasting, or do firms undo it through separations? Second, do firms comply but relegate the new women to lower positions? Third, since firms were hiring few women before, does the quota create large efficiency losses?

We answer these questions using a gender hiring quota introduced in Italy, which has one of the largest gender employment gaps in the EU.[3] Since 2021, firms that win public contracts must reserve for women at least 30 percent of the new hires needed for the project.[4] The setting is ideal for three reasons. First, the quota binds: most of the firms we study operate in construction, and before the award women account for only 9.8 percent of their hires, less than a third of the share the quota requires. Second, firms have strong incentives to comply, because the contract is worth on average 36.5 percent of their annual sales.[5] Third, the quota expires with the contract, so we can test whether its effects last. We find that the quota raises female employment at no cost to firm performance. Most new jobs, however, are temporary white-collar positions, and part of the gain fades in the years after the award.

The quota applies to public contracts funded by the National Recovery and Resilience Plan (PNRR).[6] Contracting authorities specify it ex ante in the call for tenders, so firms

[1] Employment rates refer to the population aged 25-54 (OECD, 2024).

[2] Norway introduced a 40 percent board quota in 2003. Today 28 jurisdictions have binding board quotas, and Directive (EU) 2022/2381 extends them to large listed firms across the EU.

[3] In 2024, the employment rate of women aged 25-54 was 19.2 percentage points below that of men in Italy, twice the EU average of 9.6 points (Eurostat, 2024).

[4] Existing affirmative-action programs typically set non-binding goals or target narrow groups of workers (Leonard, 1984; Miller, 2017; Szerman, 2022).

[5] Computed for the 88.5 percent of winners for which we observe balance sheets. The median is 7.0 percent.

[6] The PNRR is Italy's component of Next Generation EU, the EU's post-COVID-19 recovery program,

know whether a contract carries it before deciding to bid.[7] Waivers are hard to obtain: contracting authorities must formally justify why the quota is infeasible, for instance because the contract is small or requires fewer than three hires. By contrast, waivers are generally not accepted for labor-intensive contracts.[8] In sectors with particularly low female employment, contracting authorities may reduce the quota target, but not waive it entirely. In practice, the quota applies in full to 30 percent of PNRR contracts, which account for 47 percent of their value, is reduced in 2 percent, and is waived in 55 percent, mostly small contracts.[9] After the award, the quota binds only if the firm hires at least three workers for the project, and leaves firms free to choose how many workers to hire and in which occupations, hours, and contract types. Firms can therefore hire more women, hire fewer men, or stay below the three-hire threshold to avoid the requirement.

We build a unique dataset that links employer-employee records to the universe of Italian public procurement contracts and to firm balance sheets. Because coverage depends mainly on how a contract is funded, not on what it buys, we can compare firms awarded contracts with the quota to firms awarded comparable contracts without it, using a matched difference-in-differences design. Both groups win a public contract, so the comparison isolates the effect of the quota from that of winning. We match contracts on type of work, procedure, timing, and value, and firms on sector, size, and pre-award female employment share.[10] The key assumption is that, without the quota, two similar firms winning similar contracts would have hired and retained women in the same way.

Covered contracts are important for the firms that win them: a firm is unlikely to give up a large public contract to avoid hiring a woman. Selection into the quota is therefore likely to be a second-order concern, even though firms know before bidding whether a contract carries it. Matching on pre-award female employment and the absence of pre-trends further address this concern.

The firms in our sample are small, male-dominated, and hire infrequently. In the year before the award, the median firm employs 12 workers, one of whom is a woman, and hires

running from 2021 to 2026. PNRR contracts accounted for 8.1 percent of the value of public procurement in 2021-2025, peaking at about 18 percent in 2023. The quota is set by Article 47 of Decree-Law 77/2021, which also sets an independent 30 percent quota for workers under 36.

[7]Contracting authorities are the public entities that buy works, goods, and services, from ministries and regions to municipalities, health authorities, and publicly owned companies. They design the call for tenders, choose the award procedure, and select the winning firm.

[8]The implementing guidelines call this a "particularly stringent" burden, to be formalized before the procedure starts (Prime Minister's Decree of 7 December 2021). Labor-intensive contracts are those in which labor costs account for at least half of the contract value.

[9]The remaining 12.6 percent of contracts are unclassified.

[10]The unit of analysis is the firm-contract, and treatment is a firm's first award of a contract with the quota. Before the award, treated and control firms employ on average 20.7 and 20.9 workers, and their contracts have nearly identical values.

three new workers, none of them women. For 46 percent of firms, complying would mean hiring a woman for the first time. Women employed before the award hold mostly white-collar positions. Nearly three quarters of the firms operate in construction, where only 8.5 percent of workers are women[11] and physical strength gives men a productivity advantage (Galor and Weil, 1996; Pitt et al., 2012). Our setting is therefore a demanding test of the quota: if it works here, it is likely to work where women are easier to hire.

The quota raises female employment by 11.2 percent three years after the award, about 0.3 additional women per firm, with no change in male employment. The increase is driven by female hiring, while the probability of any hire falls by 2.8 percentage points, reflecting fewer male hires and consistent with some firms avoiding the requirement. First, the gain partly fades: female separations do not rise in the award year or the following year, but increase two years after the award. Female employment remains above its pre-award level, although the estimate becomes imprecise. Second, most additional women come from non-employment and are hired into white-collar jobs, but 81 percent of these jobs are temporary and 64 percent part-time, with no effect on wages. Third, we find no evidence of efficiency losses: revenues, assets, and profit margins do not change, and labor productivity rises by 6.7 percent, although the estimate is imprecise.

Two forces shape these responses. The first is the government's leverage as a buyer. Responses are stronger when the contract is large relative to the firm's wage bill, consistent with firms complying because they depend on public demand. The second is the combination of temporary contracts and hiring frictions, which explains why the gain fades but does not disappear. Temporary contracts can simply expire, so separations rise in the years after the award. Yet replacing a woman with a man requires a firing cost and a new hiring cost, and she accumulates skills specific to the firm. The quota thus creates a wedge between keeping the new hires and returning to the old workforce composition, even after the quota expires. Even when the jobs end, they may strengthen women's attachment to the labor market: most of these women come from non-employment, and work experience can make it easier to find the next job.

Contribution to the Literature Our paper contributes to three literatures. First, we add to the literature on gender quotas. So far, quotas have targeted positions of power, in politics and on corporate boards. They can change beliefs about women in leadership (Beaman et al., 2009, 2012) and the quality of those selected (Baltrunaite et al., 2014). In firms, evidence on performance is mixed, with some studies finding costs (Ahern and

[11] In 2024, women accounted for 8.5 percent of workers aged 25-64 employed in construction in Italy (Eurostat, 2024).

Dittmar, 2012) and others no effect (Eckbo et al., 2022; Ferrari et al., 2022), and board quotas have little effect on women below the top (Bertrand et al., 2019; Maida and Weber, 2022). Whether quotas can close the employment gap therefore remains an open question. We study a quota that acts directly on hiring, across all jobs, and examine the questions this raises: whether its effects last, which jobs women get, and whether firms bear efficiency costs.

Second, we contribute to the literature on affirmative action in employment. Existing programs fall into two groups. The first sets hiring goals for broad groups, as in the US federal contractor program, which ties affirmative action to public contracts but does not bind (Leonard, 1984; Miller, 2017; Amano-Patino et al., 2021). The second binds, but applies to the stock of employment and to narrow groups, such as persons with disabilities (Szerman, 2022). We study a quota that is binding, applies to the flow of new hires, and targets half of the population. Because it expires with the contract, we can also measure how much of the response survives once the constraint is lifted.

Finally, we contribute to the literature on public procurement as a policy tool. Governments often use their purchasing power to favor particular suppliers, and existing work studies how these preferences affect who wins contracts, procurement costs, and competition (Marion, 2007, 2009; Krasnokutskaya and Seim, 2011; Athey et al., 2013; Rosa, 2024). Conditions on what winning firms do after the award have received less attention, and the main example, the US federal contractor program, sets non-binding goals. We show that a binding condition attached to public contracts can change the hiring of private employers, and that its effect grows with the firm's reliance on public demand.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting, Section 3 the data, and Section 4 the empirical strategy. Section 5 describes the margins along which firms can respond to the quota and presents the results: the effects on female employment, the margins along which firms adjust, firm performance, and heterogeneity by the economic importance of the contract and by firm size. Section 6 concludes.

2 Institutional Setting

2.1 Public Procurement in Italy

Public procurement is one of the largest channels through which governments buy from private firms. Across OECD countries, it accounts for 12.7 percent of GDP (OECD, 2023). Italy is close to this average: public bodies spend more than €300 billion every year on works, services, and supplies.

The Italian procurement system is highly decentralized. Public calls are issued and managed by contracting authorities, including ministries, regions, municipalities, health agencies, and public companies. Each authority defines the contract object, chooses the award procedure, sets the reserve price, and selects the winner.[12]

Firms can bid alone or in a temporary consortium. The lead firm signs on behalf of all members, while all members remain jointly liable. Each contract binds the firm for a limited period, and all obligations attached to the contract apply over that period.

The contracting authority monitors execution and appoints a project manager and a works director or contract supervisor. A final inspection certifies performance before the last payment.[13]

Firms that do not comply face penalties and, in serious cases, termination. Serious breaches are recorded by ANAC and can exclude firms from future calls. For firms depending on public demand, non-compliance can therefore be costly beyond the single contract.

Winning a contract creates a labor demand shock. A firm can reallocate existing employees or hire new workers. This choice matters because the gender quota applies only to new hires.

2.2 The PNRR and the Gender Hiring Quota

The PNRR is Italy's share of NextGenerationEU. Between 2021 and 2026, it channels €191.5 billion into the Italian economy, largely through public procurement. After the common restrictions used to construct our procurement sample, PNRR funds finance 59,726 contracts worth €22.7 billion. Of these, 20,367 contracts worth €10.6 billion carry the full gender quota.

Article 47 of Decree-Law 77/2021 requires firms winning a covered contract to reserve for women at least 30 percent of the new hires needed to execute it.[14] A separate 30 percent requirement applies to workers younger than 36. We focus on the gender requirement.

The quota targets the flow of new hires, not the existing workforce. It does not prescribe occupations, wages, hours, or contract types. Like all contractual obligations, it applies only for the duration of the contract.

Contracting authorities can apply, reduce, or waive the quota, but deviations require an adequate and specific justification. Examples include small direct awards, projects expected to require fewer than three hires, specialized qualifications, and emergencies. A partial

[12]Legislative Decree 50/2016, replaced by Legislative Decree 36/2023 from July 2023.

[13]The project manager is the *Responsabile unico del progetto*; the final inspection is the *collaudo* for works and the *verifica di conformità* for services and supplies.

[14]Decree-Law 77/2021, converted into Law 108/2021. Implementing guidelines were adopted on December 7, 2021.

reduction is allowed where sectoral female employment is especially low. Across all PNRR contracts in the study years, 29.9 percent carry the full quota, 2.1 percent a reduced quota, 55.4 percent a waiver, and 12.6 percent remain unclassified.

The quota comes with possible penalties and reporting duties. Firms with 15 to 50 employees must report the gender composition of their workforce within six months. Failure to report excludes them from subsequent PNRR tenders for twelve months. Enforcement is nevertheless uncertain. We observe exposure to the quota, not compliance, and estimate the policy's effect as monitored and enforced in practice.

2.3 A Hiring Quota for the Entire Workforce

The Italian quota differs from board and political quotas because it acts on the hiring of ordinary workers (Bertrand et al., 2019; Maida and Weber, 2022; Beaman et al., 2009; Baltrunaite et al., 2014). It also differs from non-binding affirmative-action plans and stock-based disability quotas (Leonard, 1984; Miller, 2017; Szerman, 2022). It sets a binding numerical target on the flow of hires and expires with the contract.

Three features make the policy well suited to study firm responses. It binds when firms expand; quota status varies across contracts with similar economic content; and the obligation is temporary. Italy is also a setting in which the quota may matter greatly. Among eligible treated firms, women account for 12.3 percent of employees and 9.8 percent of hires, far below the 30 percent requirement.

2.4 Where the Quota Binds

The quota falls mainly on sectors with few women. Figure 1 reports sector shares among the 4,112 full-quota contracts and the female employment distribution of winning firms.

The typical winning firm employs far fewer women than the quota requires. Moreover, 68.8 percent of eligible treated firms hired at least three workers in the baseline year. Among matched treated firms with positive baseline hiring, only 9.5 percent already hired at least 30 percent women. Most firms would therefore need to alter hiring composition when the threshold is reached.

Construction firms make up 70.9 percent of all treated firms and 73.4 percent of matched treated firms. Construction is therefore a demanding test: firms may avoid hiring or place women in a narrow set of jobs. We study sectoral heterogeneity in Section 5.6.

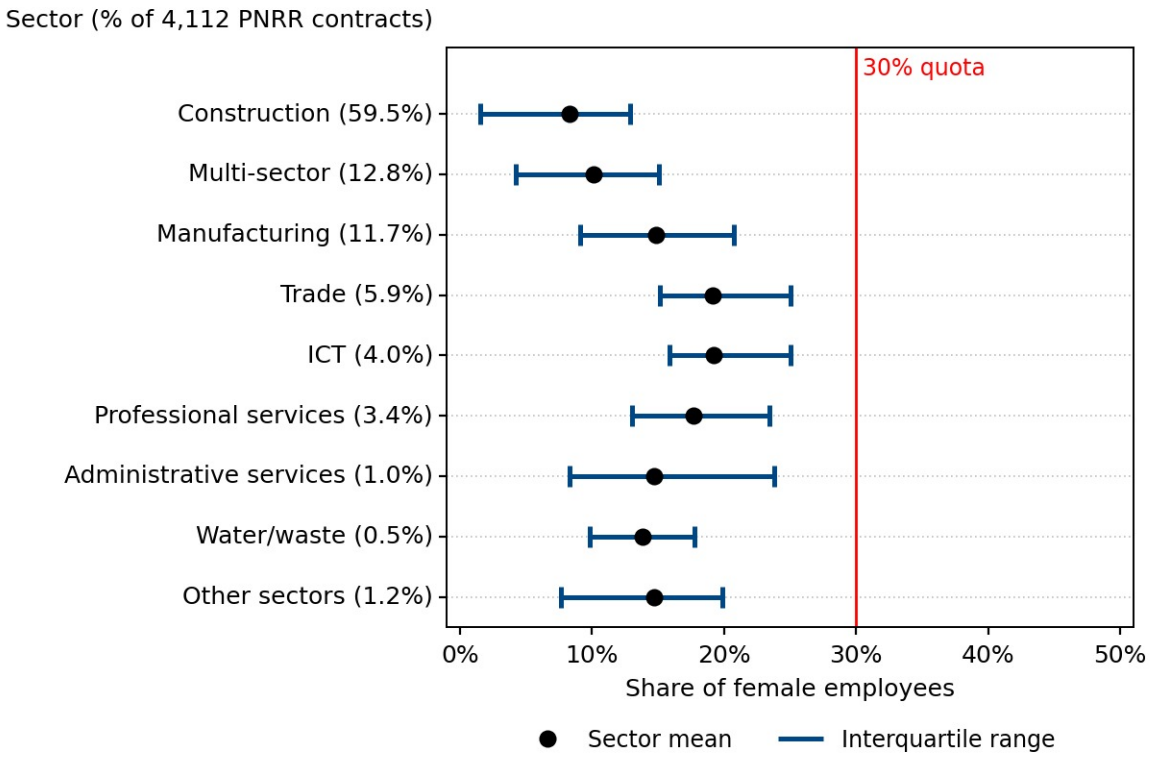


Figure 1: Exposure to the Quota by Sector

Notes: The figure reports each sector's share of the 4,112 contracts subject to the full quota and the female employment distribution among winners.

3 Data: Linking Contracts to Workers

Our data link Italian public contracts awarded from 2017 to 2025 to the full workforce of each winning firm. We combine ANAC procurement records, INPS matched employer-employee data, ReGiS quota information, and Orbis balance sheets.

3.1 Firm-Level Administrative Data

INPS records contain every private-sector employment contract from 2017 to 2025. We observe wages, weeks worked, working time, contract type, occupation, age, and gender. We construct employment, hires, separations, and their gender composition at the firm-year level. A hire is a worker employed by the firm in a year but not in the previous year. Non-employment means no private employment contract in the previous year and may include unemployment, inactivity, self-employment, or public employment.

Orbis provides sales, input costs, value added, profits, and total assets. We measure labor productivity as log value added per employee.

3.2 Public Procurement Data

ANAC records provide contract object, CPV code, procedure, location, amount, authority, funding source, and winners. ReGiS provides quota status for PNRR calls issued from 2021 to 2024. We link sources through the tender identifier.

3.3 Sample Restrictions

We focus on contracts awarded between 2021 and 2024 and worth between €15,000 and €10 million. We keep private firms with at least one baseline employee and retain only the lead firm in a consortium. The unit is the firm-contract. The final sample contains 214,073 firm-contracts won by 45,593 firms.

3.4 Descriptive Statistics

PNRR contracts represent 6.8 percent of contracts in our procurement sample and 7.3 percent of their total value. The median PNRR contract is worth €61,537, compared with €72,484 for an ordinary contract. Works account for 34.8 percent of PNRR contracts and 19.7 percent of ordinary contracts.

PNRR contracts are slightly shorter among completed contracts with observed duration. Their median duration is 4.6 months, compared with 5.5 months for ordinary contracts.

Moreover, 84.5 percent last less than one year, compared with 78.9 percent of ordinary contracts. Duration is observed for 20.6 percent of PNRR contracts and 16.8 percent of ordinary contracts.

Table 1: Public Contracts and the Gender Hiring Quota

	Ordinary	PNRR
Number of contracts	823,096	59,726
Share of contracts (%)	93.2	6.8
Total value (€ billion)	287.4	22.7
Share of total value (%)	92.7	7.3
Median value (€)	72,484	61,537
Works (%)	19.7	34.8
Duration observed (%)	16.8	20.6
Median observed duration (months)	5.5	4.6
Observed duration below one year (%)	78.9	84.5

Notes: The table uses the matching-eligible procurement universe. Duration is observed only for completed contracts.

Waivers are more common than the full quota, but the full quota covers greater value. It applies to 29.9 percent of PNRR contracts and 46.7 percent of PNRR value. Waivers apply to 55.4 percent of contracts and 18.0 percent of value. In the matching-eligible sample, the median full-quota contract is worth €95,000, compared with €47,780 for waived contracts. The most common coded waiver reason is a reduced contract amount, reported for 14.1 percent of waived contracts.

The eligible treated sample contains 4,112 firms, while the potential-control pool contains 209,961 firm-contracts won by 41,481 firms. The median eligible treated firm employs 18 workers, compared with three in the private sector. Women account for 12.3 percent of employment and 9.8 percent of hires, compared with 40.1 percent in the private sector.

Hiring is common among eligible treated firms. In the baseline year, 91.8 percent hire at least one worker and 68.8 percent hire at least three. The median firm hires five workers but no women. Female hires are less often temporary than male hires, but more often part-time and white-collar. In the matched sample, 50.8 percent of baseline female hires come from non-employment, and only 9.5 percent of firms with positive hiring already hire at least 30 percent women.

Eligible treated firms are frequent procurement participants. Between 2017 and 2020, the median treated firm won six public contracts, and 39.5 percent won at least one contract each year. Among firms appearing in the full-quota and waiver winner pools, 18.9 percent win both types.

Table 2: Baseline Characteristics of Firms Exposed to the Full Quota

	Eligible treated	Matched treated	Private sector
Firm-contracts	4,112	685	—
Median employment	18	12	3
Median hires	5	3	—
Median female hires	0	0	—
Any hire (%)	91.8	87.4	—
At least three hires (%)	68.8	55.0	—
Female employment share (%)	12.3	12.3	40.1
Female share of hires (%)	9.8	8.8	40.1
Female hiring share at least 30% (%)	—	9.5	—
Female temporary hires (%)	35.7	41.6	—
Male temporary hires (%)	58.1	59.2	—
Female part-time hires (%)	21.8	33.9	—
Male part-time hires (%)	4.9	5.6	—
Female white-collar hires (%)	89.4	88.9	—
Male white-collar hires (%)	24.1	19.1	—
Female hires from non-employment (%)	—	50.8	—

Notes: Characteristics are measured in the baseline year. The 30 percent benchmark is computed among matched treated firms with positive baseline hiring.

Quota contracts are economically important. Among eligible treated firms with balance-sheet data, the first quota contract is worth on average 36.5 percent of annual sales and 7.0 percent at the median. The median contract equals 0.62 times annual wage bill. For 25 percent of firms, it exceeds 28.7 percent of sales, and for 10 percent it exceeds 84.4 percent. Sales are observed for 88.5 percent of eligible treated firms.

4 Empirical Strategy

4.1 Comparing Winners of Similar Contracts

The main empirical challenge is selection into contracts with the quota. Contracting authorities decide quota status before publication, and firms observe it before bidding. A simple comparison would therefore mix the quota's effect with differences in firms, sectors, authorities, and projects.

We compare firms winning nearly identical contracts, with and without the quota. Every firm in the comparison wins a public contract and therefore faces a similar demand shock. The central difference is that women must make up at least 30 percent of any new hires needed by the treated firm.

4.2 Treatment and Comparison Groups

Treatment is defined at the firm level. For each firm, we keep its first PNRR contract subject to the full quota. This leaves 4,112 treated firm-contracts.

Control firm-contracts carry no quota. They include ordinary contracts and PNRR contracts with a complete waiver. We exclude contracts with a reduced quota. Firms that ever win a full-quota contract cannot serve as controls. The potential-control pool contains 209,961 firm-contracts won by 41,481 firms.

The same control firm can appear more than once because it may win several eligible contracts and because matching is with replacement. We account for this repetition in the standard errors.

4.3 Matching on Contracts and Firms

We match each treated firm-contract to one control firm-contract in two steps. First, contracts must have the same object and award procedure, be awarded within six months, and share a CPV code. We first search at four digits, then three, and then two.[15] Contract values must differ by less than 10 log points and €1,500.

Second, firms must operate in the same macro-sector. Log baseline employment must differ by less than 0.30, and the female employment share by less than 10 percentage points. Among candidates, we select the nearest control using standardized Euclidean distance.

The procedure yields 685 matched pairs, or 16.7 percent of treated firm-contracts. The 685 controls correspond to 665 distinct firm-contracts won by 642 firms. Table 3 shows that treated and control firms are nearly identical before the award. The average firm employs about 21 workers, of whom 2.6 are women, and wins a contract worth about €92,000.

[15] In the matched sample, 73 percent of pairs share a four-digit CPV code, 13 percent a three-digit code, and 14 percent a two-digit code.

Table 3: Baseline balance between all treated firms, matched treated firms, and matched controls

	All treated		Matched treated		Matched controls	
Variable	Mean	Median	Mean	Median	Mean	Median
Panel A: Variables used in matching						
Female employment share	0.11	0.10	0.11	0.11	0.11	0.11
Total employment (log)	3.10	2.89	2.63	2.48	2.63	2.48
Industry	0.13		0.13		0.13	
Services	0.16		0.13		0.13	
Construction	0.59		0.70		0.70	
Agriculture	0.00		0.00		0.00	
Multisector	0.12		0.04		0.04	
Procurement amount	836,793.47	140,667.52	92,790.81	47,638.07	92,799.21	47,607.00
Panel B: Main baseline outcomes						
Total employment	43.63	18.00	21.39	12.00	21.51	12.00
Female employment	5.39	2.00	2.63	1.00	2.60	1.00
Male employment	38.24	17.00	18.75	11.00	18.91	11.00
Blue-collar employment	25.17	12.00	13.73	9.00	14.32	9.00
White-collar employment	18.46	4.00	7.66	3.00	7.19	2.00
Weeks worked	1,684.72	653.00	839.24	463.00	834.05	468.00
Wage bill	1,207,789.97	367,138.50	556,132.69	251,897.00	526,392.38	252,418.00
White-collar employment share	0.32	0.22	0.27	0.20	0.25	0.18
Female white-collar employment share	0.10	0.09	0.11	0.10	0.10	0.10
Total hires	11.09	5.00	4.95	3.00	5.12	3.00
Female hires	1.08	0.00	0.44	0.00	0.45	0.00
Average weekly wage	576.79	524.18	545.12	508.44	546.84	510.56
Average weekly wage female	598.14	543.95	561.14	528.10	564.49	524.83
Average weekly wage (log)	6.25	6.23	6.21	6.20	6.22	6.20
Gender wage gap (%)	-1.44	-0.59	-1.94	-1.74	-2.79	-1.98
N. observations	4,112		685		685	
N. firm-events	4,112		685		665	
N. unique firms	4,112		685		642	

Notes: Statistics are measured in the baseline year. "All treated" reports eligible treated firm-events before matching; "Matched treated" and "Matched controls" report the matched firm-events used in the estimation. A control firm matched to different procurement events enters once for each event, possibly with different baseline years. Panel A reports the firm-level matching variables and Panel B selected baseline outcomes. The female share of white-collar employment is computed for firms with at least one white-collar employee, and the average female weekly wage for firms with at least one female employee. The gender wage gap is the log difference in firm-level FTE weekly wages between men and women, read as the percentage by which men's wages exceed women's, and is computed for firms with at least one worker of each gender.

The matched sample differs from all treated firms. Matched firms have median employment of 12 workers, compared with 18 among all treated firms. Their median contract is about €48,000, compared with €141,000. Construction accounts for 73.4 percent of matched treated firms and 70.9 percent of all treated firms. Baseline female employment shares are approximately 11 percent in both groups.

4.4 Estimating Equation

We estimate a difference-in-differences model within matched pairs. For firm i, contract c, pair p, and year t:

$$Y_{icpt} = \beta \left(\text{Treated}_{icp} \times \text{Post}_{ict}\right) + \alpha_{ic} + \lambda_{pt} + \varepsilon_{icpt}. \tag{1}$$

Firm-by-contract fixed effects, α_{ic}, absorb time-invariant differences. Pair-by-year fixed effects, λ_{pt}, absorb shocks common to each pair. The coefficient β measures the average quota effect on treated firms in the matched sample. We cluster standard errors at the firm level.

We also estimate:

$$Y_{icpt} = \sum_{k \neq -1} \beta_k \left(\text{Treated}_{icp} \times \mathbf{1}[t - E_{ic} = k]\right) + \alpha_{ic} + \lambda_{pt} + \varepsilon_{icpt}, \tag{2}$$

where the year before the award is the reference period. Pre-award coefficients test for differential trends.

4.5 Identifying Assumption and Threats

The identifying assumption is that treated firms and matched controls would have followed parallel paths without the quota. Matching makes this assumption more plausible, and event studies show no evidence of differential pre-trends.

Four concerns remain. Firms may select into quota calls on unobserved characteristics. The concurrent youth quota may also affect hiring. Treated and control firms may compete in the same labor markets. Finally, we observe exposure rather than compliance. Our estimates therefore identify the effect of the quota as implemented in practice.

5 How Firms Can Respond to a Hiring Quota

The quota can affect whether firms hire, how many women and men they hire, and the jobs offered to women. These margins are not mutually exclusive. All estimates are relative to

matched controls, so they isolate the quota's differential effect from the common effect of winning a contract.

5.1 The Effect on Female Hiring and Female Employment

The quota raises female hiring. Treated firms hire 0.11 more women per year than controls, an increase of 25.9 percent over a baseline of 0.44. The increase is concentrated in the award year and the following year.

Female employment rises by 11.2 percent after the award, while male employment does not change. Before the award, treated and control firms follow similar paths. The quota therefore changes whom firms hire and increases the stock of female employment without displacing men.

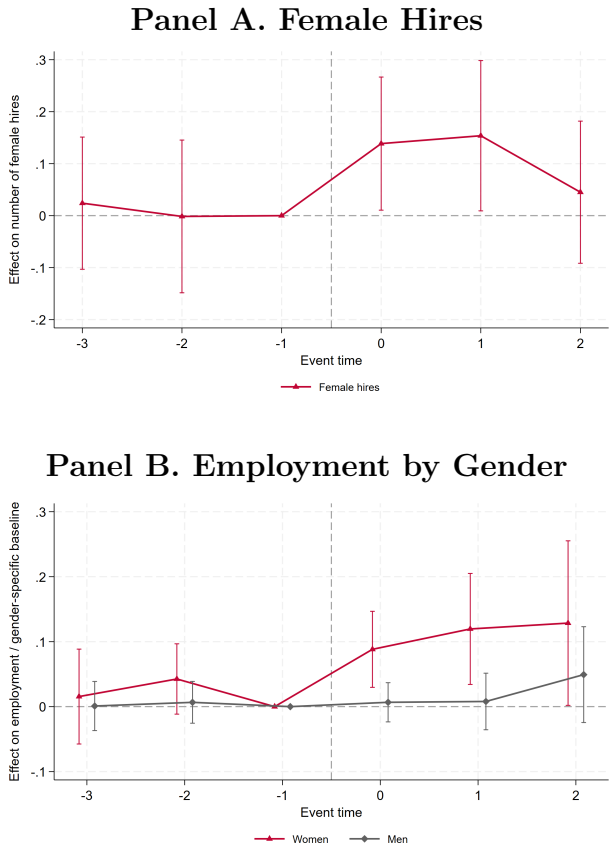


Figure 2: Female Hires and Employment by Gender

Notes: Event-study coefficients compare treated firms with matched controls. Event time zero is the award year; the preceding year is omitted. Bars show 95 percent confidence intervals with standard errors clustered at firm level.

5.2 Extensive Margin: Whether to Hire

A firm must comply only if it hires at least three project workers. It can therefore avoid the constraint by reallocating existing workers or remaining below the threshold.

The probability of any hire falls by 2.8 percentage points from a baseline of 87 percent. The probability of hiring a man falls by 3.0 percentage points, while the probability of hiring a woman does not change. This pattern is consistent with some firms avoiding constrained recruitment.

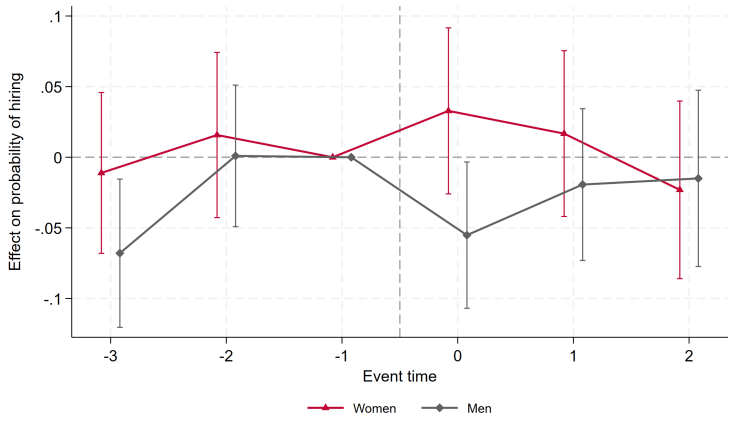


Figure 3: Probability of Hiring, by Gender

Notes: Event-study coefficients compare treated firms with matched controls. The outcome is an indicator for hiring at least one worker of each gender.

5.3 Intensive Margin: How Many Women and Men to Hire

Firms can comply by expanding female hiring or substituting women for men. The female hiring rate rises by 0.7 percentage points, or 35.8 percent of its baseline value of 2.0 percent.

The employment gains partly fade after the quota expires. Female separations do not rise in the award year or the following year, but increase two years after the award. Female employment remains above its pre-award level, although the estimate becomes imprecise.

5.4 Compositional Margin: Which Women, in Which Jobs, and on What Terms

Most additional women come from non-employment. Hires from non-employment account for 26.8 percentage points of the 35.8 percent increase in the female hiring rate, and hires from other firms for 9.0 points. About 75 percent of additional female hires were therefore not employed before.

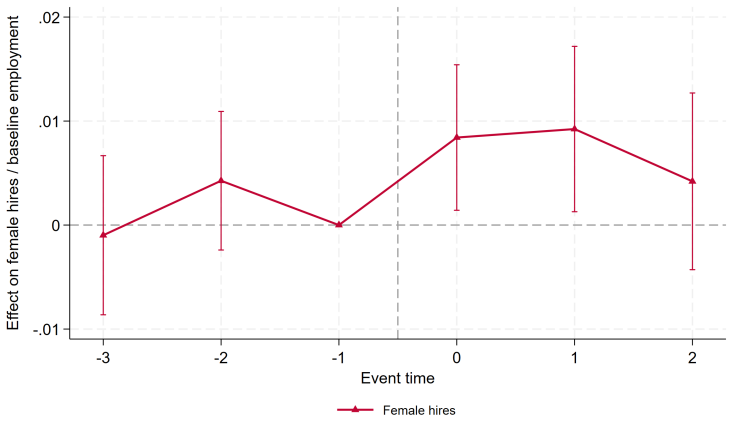


Figure 4: Female Hiring Rate

Notes: The outcome is female hires divided by baseline employment.

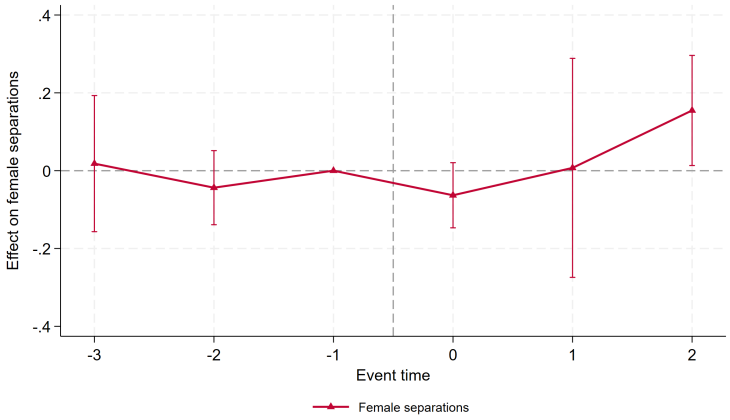


Figure 5: Female Separations

Notes: The outcome is the number of women employed in the previous year who are no longer employed by the firm.

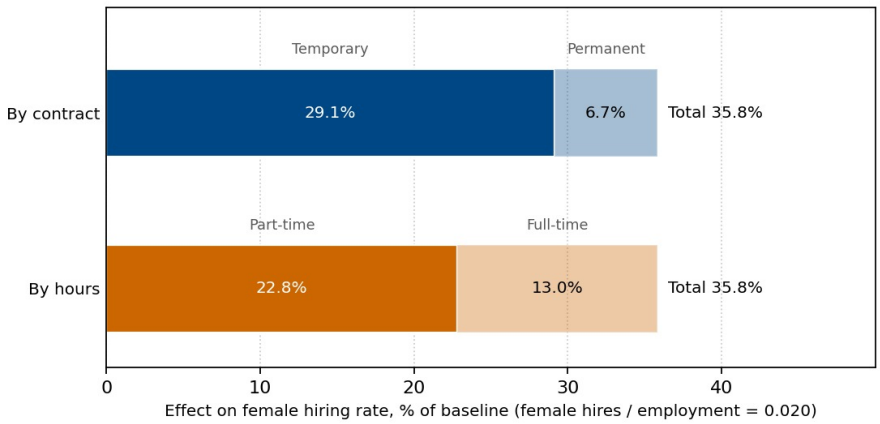


Figure 6: Composition of Additional Female Hires

Notes: The figure decomposes the post-award increase in the female hiring rate by previous labor-market status and occupation.

The additional women mainly enter white-collar positions. White-collar hires account for 26.3 percentage points of the increase and blue-collar hires for 9.5 points. The new jobs are also mostly temporary and part-time. Temporary contracts account for 81 percent of additional hires and part-time contracts for 64 percent.

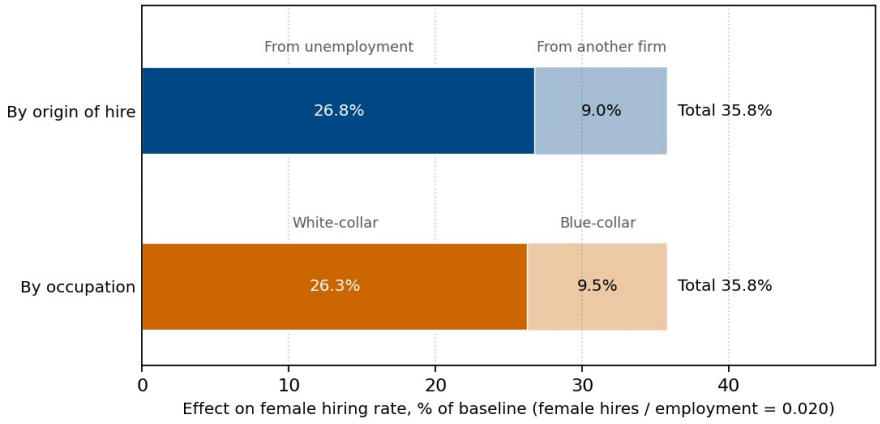


Figure 7: Terms of Employment of Additional Female Hires

Notes: The figure decomposes the increase in the female hiring rate by contract type and working time.

Weekly wages do not change. Estimates for women and men, new hires, and white-collar workers are close to zero.

5.5 Firm Performance

The quota does not harm firm performance. Labor productivity rises by 6.7 percent after the award. Estimates for revenues, total assets, and profit margins are positive but small

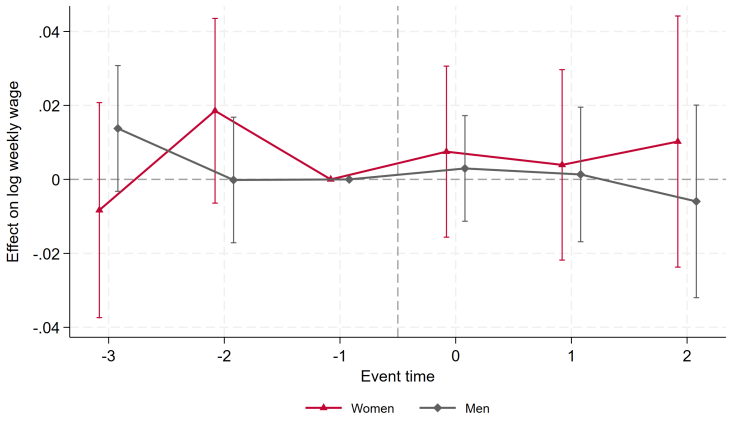


Figure 8: Weekly Wages by Gender

and not statistically different from zero. We interpret the productivity estimate cautiously because many additional women work part-time and the confidence interval is wide.

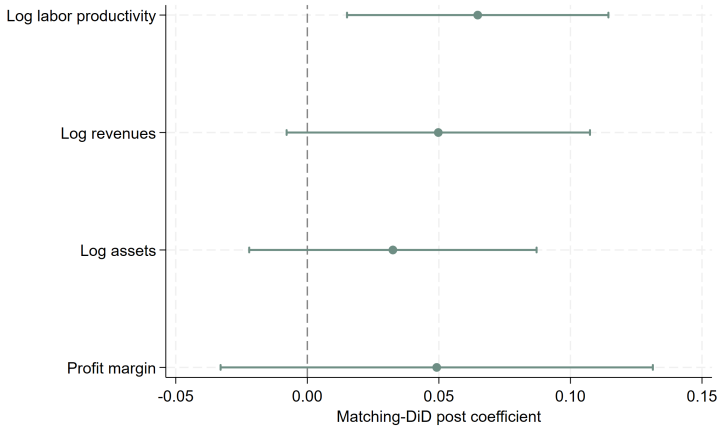


Figure 9: Firm Performance

5.6 Heterogeneity

Responses are concentrated where the contract matters most. We measure importance as contract value relative to the treated firm's average annual wage bill in the two years before the award. In the top tercile, the probability of any hire falls and the female employment share rises. Effects in the bottom tercile are not distinguishable from zero.

Small firms drive employment gains, while hiring avoidance is concentrated among mid-sized firms. In small firms, female weeks worked and female hiring rise sharply. Mid-sized firms reduce hiring, and large firms show no significant response.

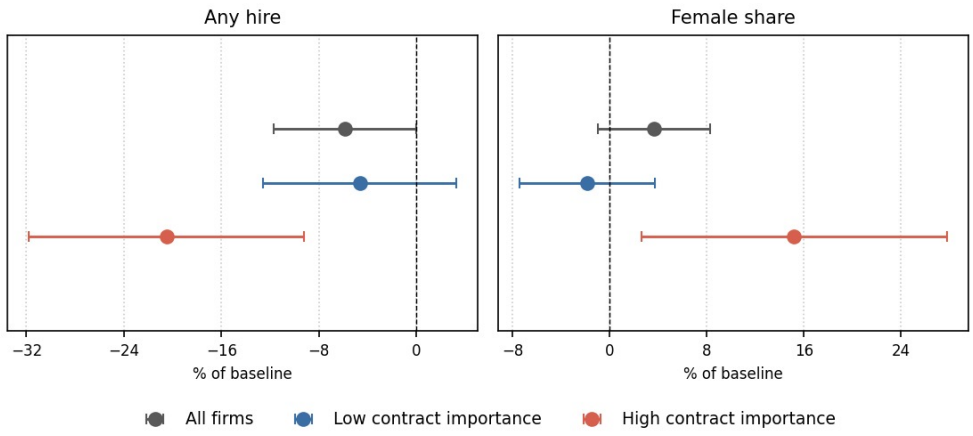


Figure 10: Heterogeneity by Contract Importance

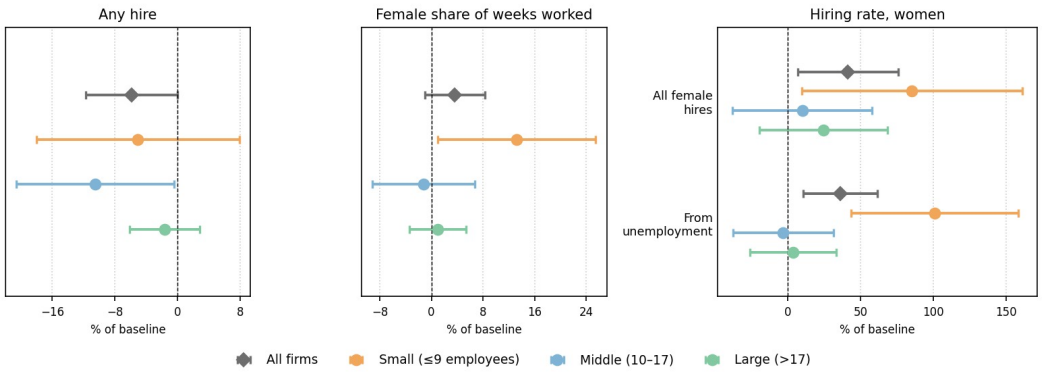


Figure 11: Heterogeneity by Firm Size

6 Conclusions

We study a gender quota on new hires attached to public contracts in Italy. Firms exposed to the quota employ more women than comparable firms winning similar contracts without it, with no change in male employment and no evidence of efficiency losses. Most additional women come from non-employment and enter white-collar jobs, but these jobs are mostly temporary and part-time. Some firms avoid the requirement by hiring less, and part of the employment gain fades after the contract ends.

Three lessons follow. Binding conditions attached to public contracts can change private employers' hiring, and their reach grows with firms' reliance on public demand. Temporary quotas can have effects that outlast the formal obligation, although temporary contracts make the gains less persistent. Finally, even short jobs may strengthen women's attachment to the labor market. Studying the later careers of women hired under the quota is a natural next step.

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A Matching and Sample

Table A1: Matching Criteria

Criterion	Rule
Contracts	
Contract object	Same object: works, services, or supplies
Award procedure	Same procedure
Timing	Award within six months
Product classification	Same CPV at four, three, or two digits
Contract value	Log difference below 0.10 and absolute difference below €1,500
Firms	
Sector	Same macro-sector
Size	Log-employment difference below 0.30
Female share	Difference below 10 percentage points
Selection	Nearest standardized Euclidean distance, with replacement

Table A2: Matched and All Treated Firms

	Matched treated	All treated
Employment (median)	12	18
Female share of employment	0.111	0.109
Contract value (median, euros)	47,638	140,668
Construction share	0.734	0.709
Firm-contracts	685	4,112

Notes: The table compares matched treated firm-contracts with all eligible treated firm-contracts. Firm characteristics are measured in the baseline year.

B Employment Outcome and the Female Share

For firm i, let F_{it} and M_{it} denote female and male employment, and let $N_{i,-1}$ denote baseline employment. Define

$$Y_{it}^{F} = \frac{F_{it}}{N_{i,-1}\overline{s}_{-1}^{F}}, \qquad Y_{it}^{M} = \frac{M_{it}}{N_{i,-1}\overline{s}_{-1}^{M}}.$$

For small changes in the female share $s = F/(F+M)$:

$$\Delta s \approx \overline{s}_{-1}^{F}\overline{s}_{-1}^{M}\left(\beta^{F} - \beta^{M}\right).$$

The female employment share therefore rises when the treatment effect on scaled female employment exceeds the effect on scaled male employment.